PROJECT TITLE:

INTELLIGENT GPA PREDICTION AND ACADEMIC DECISION SUPPORT SYSTEM

PROJECT REPORT

COURSE: FUNDAMENTALS OF AI AND ML

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Abstract

This project focuses on predicting GPA based on expected grades and assisting students in making effective academic decisions. Many students face difficulty in deciding which subject to prioritize for improving their performance. The system analyzes subjects using credits, difficulty, and preparation level to calculate a priority score. Based on this, it recommends the most effective subject to focus on. The project uses simple AI-inspired decision-making logic to provide a practical and easy-to-use academic planning tool.

PROBLEM STATEMENT

STUDENTS OFTEN STRUGGLE TO DECIDE WHICH SUBJECTS REQUIRE MORE ATTENTION TO IMPROVE THEIR GPA. THERE IS NO STRUCTURED METHOD AVAILABLE TO GUIDE THEM IN MAKING EFFECTIVE STUDY DECISIONS, WHICH LEADS TO INEFFICIENT PREPARATION AND POOR TIME MANAGEMENT.

WHY THIS PROBLEM MATTERS

THIS PROBLEM IS COMMON AMONG STUDENTS IN REAL ACADEMIC ENVIRONMENTS. BEFORE EXAMS, STUDENTS ARE OFTEN CONFUSED ABOUT WHERE TO FOCUS THEIR EFFORTS. SOLVING THIS PROBLEM HELPS STUDENTS MAKE BETTER USE OF THEIR TIME AND IMPROVES THEIR ACADEMIC PERFORMANCE. THIS MAKES THE PROJECT RELEVANT AND PRACTICAL IN REAL-LIFE SITUATIONS

APPROACH TO SOLUTION

THE SOLUTION IS BASED ON A SIMPLE RULE-BASED DECISION SYSTEM:

THE USER INPUTS SUBJECT DETAILS

GPA IS CALCULATED USING WEIGHTED AVERAGE

EACH SUBJECT IS EVALUATED USING AN IMPACT VS EFFORT MODEL

A SCORE IS ASSIGNED TO EACH SUBJECT

THE SYSTEM SUGGESTS THE BEST SUBJECT TO IMPROVE

THIS APPROACH APPLIES CONCEPTS OF DECISION-MAKING AND PROBLEM-SOLVING FROM AI.

CONCEPT USE

This project applies several fundamental concepts from Artificial Intelligence, Machine Learning, and Programming:

1. Problem Solving Approach
   The system identifies a real-world problem (academic decision-making) and provides a structured solution using logical steps.

2. Decision-Making System
   The program uses rule-based decision logic to recommend the best subject for improving GPA.

3. Heuristic Method
   A scoring function (impact vs effort) is used as a heuristic to prioritize subjects efficiently.

4. Weighted Average (Mathematics)
   GPA is calculated using a weighted average formula based on credits and grade points.

5. Data Representation
   Subjects are stored using lists and dictionaries, representing structured data.

6. Iterative Algorithms
   Loops are used to process multiple subjects and a improvements.

7. Simulation Technique
   The system simulates grade improvement (to "S") to predict GPA changes.

8. Conditional Logic
   IF-ELSE conditions are used for recommendations and performance evaluation.

9. User Input Handling
   The system interacts with users through a command-line interface.

10. Basic AI Concept (Rational Decision Making)
    The system selects the best action (subject to improve) based on maximizing GPA improvement.

KEY DESIGN DECISIONS

- USED A SCORING SYSTEM (IMPACT VS EFFORT) INSTEAD OF RANDOM SUGGESTION
- USED WEIGHTED GPA CALCULATION FOR ACCURACY
- CHOSE COMMAND-LINE INTERFACE FOR SIMPLICITY AND EXECUTION
- USED GRADE IMPROVEMENT SIMULATION (TO S) FOR PREDICTION

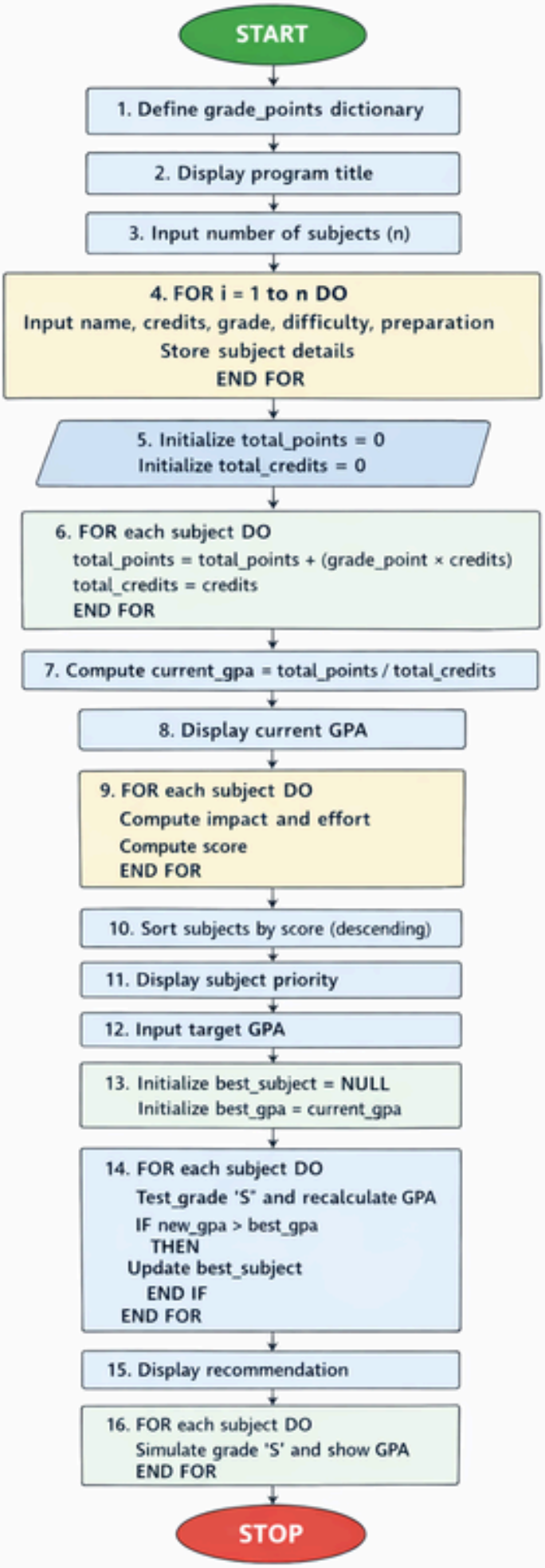
THESE DECISIONS HELPED KEEP THE SYSTEM SIMPLE YET EFFECTIVE.

METHODOLOGY

1. TAKE SUBJECT DETAILS AS INPUT
2. CALCULATE PREDICTED GPA
3. COMPUTE SUBJECT PRIORITY SCORE
4. RANK SUBJECTS
5. SUGGEST BEST SUBJECT
6. PERFORM GPA PREDICTION ANALYSIS

ALGORITM

Algorithm for GPA Prediction and Decision Agent



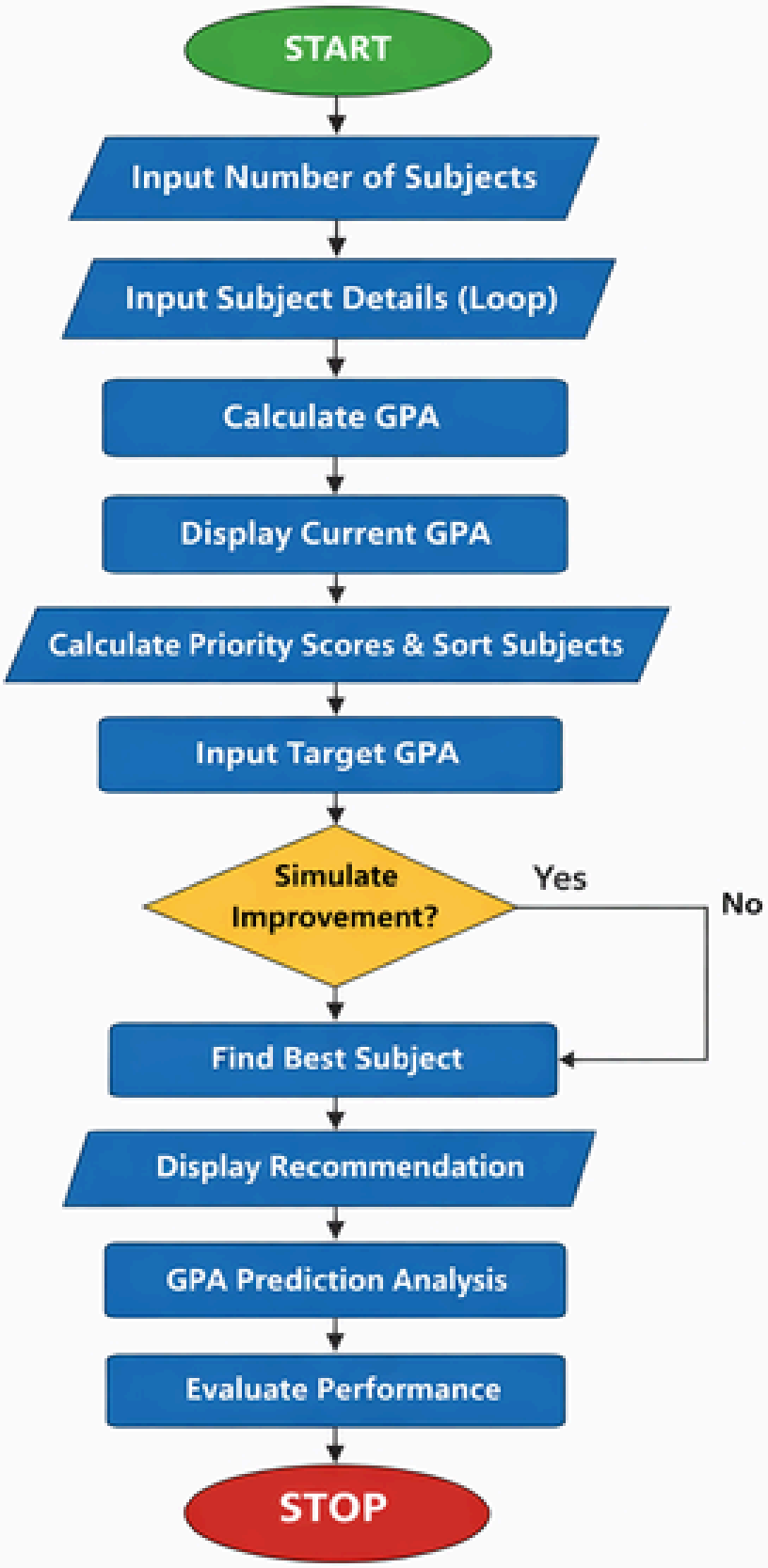
PSEUDO CODE

```
START

DEFINE GRADEPOINTS DICTIONARY

PRINT TITLE

INPUT NUMBER OF SUBJECTS (N)

CREATE EMPTY LIST SUBJECTS

FOR I FROM 1 TO N:
    INPUT SUBJECT NAME
    INPUT CREDITS
    INPUT EXPECTED GRADE
    INPUT DIFFICULTY
    INPUT PREPARATION LEVEL

    STORE ALL VALUES IN SUBJECTS LIST

INITIALIZE TOTALPOINTS = 0
INITIALIZE TOTALCREDITS = 0

FOR EACH SUBJECT IN SUBJECTS:
    TOTALPOINTS += GRADEPOINT × CREDITS
    TOTALCREDITS += CREDITS

CURRENTGPA = TOTALPOINTS / TOTALCREDITS

DISPLAY CURRENT GPA

FOR EACH SUBJECT:
    CALCULATE IMPACT = CREDITS × (10 – GRADEPOINT)
    CALCULATE EFFORT = DIFFICULTY + (5 – PREPARATION)
    SCORE = IMPACT / (EFFORT + 1)

SORT SUBJECTS BASED ON SCORE (DESCENDING)

DISPLAY SUBJECT PRIORITY

INPUT TARGET GPA

INITIALIZE:
    BESTSUBJECT = NONE
    BESTGPA = CURRENTGPA
```

```
FOR EACH SUBJECT:
    STORE ORIGINAL GRADE
    SET GRADE = "S"

    RECALCULATE GPA

    IF NEW GPA > BEST GPA:
        UPDATE BEST GPA
        UPDATE BEST SUBJECT

    RESTORE ORIGINAL GRADE

DISPLAY RECOMMENDATION

FOR EACH SUBJECT:
    TEMPORARILY SET GRADE = "S"
    RECALCULATE GPA
    DISPLAY GPA
    RESTORE GRADE

CHECK PERFORMANCE BASED ON GPA:
    IF >= 9 → EXCELLENT
    ELSE IF >= 8 → VERY GOOD
    ELSE IF >= 7 → GOOD
    ELSE → NEEDS IMPROVEMENT

DISPLAY NOTE

END
```

FLOWCHART

Intelligent GPA Prediction and Decision Agent



TOOLS AND TECHNOLOGIES USED

PYTHON

COMMAND-LINE INTERFACE

LISTS AND DICTIONARIES

IMPLEMENTATION DETAILS

THE PROJECT IS IMPLEMENTED IN PYTHON USING BASIC PROGRAMMING CONCEPTS. GPA IS CALCULATED USING A WEIGHTED AVERAGE FORMULA. A SCORING MECHANISM IS USED TO EVALUATE SUBJECTS. THE SYSTEM SIMULATES IMPROVEMENT BY ASSUMING THE BEST GRADE AND RECALCULATES GPA TO PROVIDE RECOMMENDATIONS.

How to Run the Program

To run this project, follow the steps below:

1. Install Python (version 3.x) on your system.
2. Download or clone the project repository from GitHub.
3. Open terminal / command prompt
4. Navigate to the project folder using:

cd Intelligent-GPA-Prediction-and-Academic-Decision-Support-System

5. Run the program using:

gpa_decision_agent.py

6. Enter the required inputs:

- Number of subjects
- Subject details (name, credits, grade, difficulty, preparation)
- Target GPA

7. The system will display:

- Predicted GPA
- Subject priority
- Recommendation
- GPA analysis
- Performance result

PROGRAM

```python
# Intelligent GPA Prediction and Academic Decision Agent

grade_points = {
    "S": 10, "A": 9, "B": 8, "C": 7,
    "D": 6, "E": 5, "F": 0
}

print("===== GPA Prediction and Decision Agent =====\n")

subjects = []

n = int(input("Enter number of subjects: "))

for i in range(n):
    print("\nSubject", i + 1)

    name = input("Subject name: ")
    credits = float(input("Credits: "))
    grade = input("Expected Grade (S,A,B,C,D,E,F): ").upper()
    difficulty = int(input("Difficulty (1-5): "))
    prep = int(input("Preparation Level (1-5): "))

    subjects.append({
        "name": name,
        "credits": credits,
        "grade": grade,
        "difficulty": difficulty,
        "prep": prep
    })

total_points = 0
```

Ln: 1 Col: 0

OUTPUT

```
Python 3.13.9 (v3.13.9:8183fa5e3f7, Oct 14 2025, 10:27:13) [Clang 16.0.0 (clang-1600.0.26.6)] on dar
win
Enter "help" below or click "Help" above for more information.
>>>
============ RESTART: /Users/mdjahiruddinahmed/Documents/vity pro.py ===========
===== GPA Prediction and Decision Agent =====

Enter number of subjects: 5

Subject 1
Subject name: CHEMISTRY
Credits: 4
Expected Grade (S,A,B,C,D,E,F): A
Difficulty (1-5): 4
Preparation Level (1-5): 5

Subject 2
Subject name: CALCULUS
Credits: 4
Expected Grade (S,A,B,C,D,E,F): B
Difficulty (1-5): 5
Preparation Level (1-5): 3

Subject 3
Subject name: CSE
Credits: 4
Expected Grade (S,A,B,C,D,E,F): A
Difficulty (1-5): 3
Preparation Level (1-5): 5

Subject 4
Subject name: ENGLISH
Credits: 2
```

```
Credits: 2
Expected Grade (S,A,B,C,D,E,F): A
Difficulty (1-5): 1
Preparation Level (1-5): 5

Subject 5
Subject name: EEE
Credits: 4
Expected Grade (S,A,B,C,D,E,F): A
Difficulty (1-5): 5
Preparation Level (1-5): 5

Predicted Current GPA: 8.78

===== SUBJECT PRIORITY =====
- CALCULUS | Score: 1.0
- CSE | Score: 1.0
- ENGLISH | Score: 1.0
- CHEMISTRY | Score: 0.8
- EEE | Score: 0.67

Enter target GPA: 9.8

Target GPA: 9.8

===== RECOMMENDATION =====

Best way to reach your target GPA (9.8):

Focus on: CALCULUS
Suggestion: Practice concepts regularly

Predicted GPA: 9.22
```

```
===== SUBJECT PRIORITY =====
- CALCULUS | Score: 1.0
- CSE | Score: 1.0
- ENGLISH | Score: 1.0
- CHEMISTRY | Score: 0.8
- EEE | Score: 0.67

Enter target GPA: 9.8

Target GPA: 9.8

===== RECOMMENDATION =====

Best way to reach your target GPA (9.8):

Focus on: CALCULUS
Suggestion: Practice concepts regularly

Predicted GPA: 9.22
Effort score: 7

===== GPA PREDICTION ANALYSIS =====
If CHEMISTRY gets S → GPA: 9.0
If CALCULUS gets S → GPA: 9.22
If CSE gets S → GPA: 9.0
If ENGLISH gets S → GPA: 8.89
If EEE gets S → GPA: 9.0

===== PERFORMANCE =====
Very Good

Note: Predictions are based on input and may vary in real exams.
>>>
```

Ln: 78 Col: 0

CHALLENGES FACED

DESIGNING A FAIR SCORING FORMULA

BALANCING IMPACT AND EFFORT

HANDLING USER INPUTS CORRECTLY

MAKING OUTPUT EASY TO UNDERSTAND

RESULTS

THE SYSTEM SUCCESSFULLY PREDICTS GPA AND PROVIDES A RANKED LIST OF SUBJECTS. IT HELPS USERS IDENTIFY HIGH-IMPACT SUBJECTS AND IMPROVES DECISION-MAKING FOR STUDY PLANNING.

LIMITATIONS

- RULE-BASED (NOT A TRAINED ML MODEL)
- DEPENDS ON USER INPUT
- ASSUMES IDEAL IMPROVEMENT
- DOES NOT INCLUDE REAL-WORLD UNCERTAINTY

WHAT I LEARNED

HOW TO APPLY AI CONCEPTS LIKE DECISION-MAKING

IMPORTANCE OF PROBLEM-SOLVING APPROACH

DESIGNING LOGIC FOR REAL-WORLD PROBLEMS

IMPROVING CODE STRUCTURE AND CLARITY

Future Improvements

Add machine learning model

Build GUI/web version

Use real academic data

Improve prediction accuracy

Conclusion

This project demonstrates how simple AI-inspired logic can be used to solve real academic problems. It helps students plan their studies effectively by predicting GPA and suggesting improvements. The system is simple, practical, and useful for real-world applications.